

Human Resource Management and AI: Building A Collaborative Contextual Transparency Database

ELLEN SIMPSON, Communications, University of Alaska Southeast, USA

RYAN ERMOVICK, Data Science, University of Virginia, USA

MONA SLOANE, Data Science & Media Studies, University of Virginia, USA

While calls for AI transparency are growing, most approaches still treat transparency as a technical property detached from the contexts in which systems are used. We argue instead that AI transparency emerges through social and technologically-situated professional practice. To study this, we develop a database-driven method for contextual AI transparency in recruiting, drawing on 100+ interviews with HR practitioners and iterative practitioner evaluations to construct a database of HR technologies capturing product functionality, claims, assumptions, and AI clarity. Our findings show that transparency is not a fixed system attribute, but a relational outcome shaped by how tools are represented and used in practice. By examining gaps between claims, assumptions, and functionality, we surface forms of opacity overlooked by technical notions of explainability. We contribute a database-building method for analysis of AI systems, empirical insights into transparency in HR technologies, and design implications for practice-centered approaches to AI transparency.

CCS Concepts: • **Human-centered computing** → **Computer supported cooperative work**; **Empirical studies in collaborative and social computing**; *Collaborative and social computing systems and tools*.

Additional Key Words and Phrases: Human Resources, Human Resources Technologies, Database Building, Contextual AI Transparency

ACM Reference Format:

Ellen Simpson, Ryan Ermovick, and Mona Sloane. 2026. Human Resource Management and AI: Building A Collaborative Contextual Transparency Database. In *Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO '26)*, November 05–07, 2026, Munich, Germany. ACM, New York, NY, USA, 19 pages. <https://doi.org/10.1145/3846167.3848594>

1 Introduction

Artificial intelligence (AI) tools are increasingly embedded in human resource technologies (HR tech) that support job candidate recruitment, from candidate sourcing and screening to interviewing and candidate selection. These systems mediate access to the labor market, and create an ecosystem of human and machine decision-making in hiring decisions [9, 41, 66]. Although HR tech products and vendors promise increased recruiter efficiency and improved hiring outcomes[56], their technical black boxes and the difficulty of reverse-engineering their outputs [52, 70] can introduce inefficiency, bias, and other harms into hiring [7, 9, 40, 47]. Consequently, there is increased demand for AI transparency, such as clarity about how AI models operate and shape human decision-making *in context* [20, 34, 67, 69].

Authors' Contact Information: Ellen Simpson, easimpson2@alaska.edu, Communications, University of Alaska Southeast, Ketchikan, Alaska, USA; Ryan Ermovick, Data Science, University of Virginia, Charlottesville, Virginia, USA; Mona Sloane, mona.sloane@virginia.edu, Data Science & Media Studies, University of Virginia, Charlottesville, Virginia, USA.



This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

EAAMO '26, Munich, Germany

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-3015-3/2026/11

<https://doi.org/10.1145/3846167.3848594>

Common approaches to AI transparency tend to conceptualize transparency as a technical property of AI systems, focusing on interpretability and explainability of model outputs [18, 39]. This approach assumes that more interpretable systems necessarily produce meaningful transparency for end-users within specific contexts [20, 46] (e.g., in recruiting). Yet recruiting and hiring decisions emerge through interactions between multiple tools, actors, and processes, rather than a single system [9, 60, 65, 68]. As a result, while an AI system might be technically transparent, the model may still remain opaque to recruiters when its explanations do not address their informational needs or fit their contextual decision-making processes [11, 14, 50, 51, 61, 66].

This challenge is compounded by a fragmented HR tech ecosystem where interconnected human and machine decisions unfold across distributed platforms and tools, each with its own data structures, assumptions and claims about AI functionality [6, 68]. Efforts to improve transparency often remain highly technical and difficult to translate into professional contexts, limiting their usefulness for practitioners [20, 46, 67]. Regulatory requirements for hiring further heighten the need for AI transparency that is useful in everyday recruiting practice [1, 3, 24, 73]. In this paper, therefore, we treat transparency as a relational outcome of professional practice, rather than a fixed attribute of an individual AI model or HR tech product.

We call this social, technical, and relational outcome of professional practice **contextual transparency** [67]. In recruiting, AI transparency emerges not only in how these systems and tools function, but also in how they are described to and interpreted by their users, and subsequently integrated into existing professional practice. This aligns with recent research on contextual and social forms of transparency, which emphasize that meaningful understanding of AI systems and their outputs depend on the context of their use in specific social and professional practices [14, 20, 30, 67].

To study contextual transparency within job candidate recruitment, we developed the TARAI Index¹, a database of HR tech drawn from over 100 practitioner interviews and iterative participatory evaluation. The database documents product functionality in comparison to vendor and designer claims, explores embedded assumptions and assesses AI clarity—how the products examined represent and use AI—so that HR tech products can be compared across the hiring ecosystem. The TARAI Index is designed for recruiters and other HR practitioners who must interpret and work within these systems, and those involved in enterprise-level procurement and implementation of these systems within organizations who may be able to act on comparative information. While both of these groups need contextual transparency, the power to do something about a *lack* of it is very different. Recruiters use and manage HR tech that they do not choose and cannot independently replace, while those in technology procurement often do not have a full understanding of the day-to-day workplace practices of the intended end-users of the products they acquire. The database we developed supports practitioners in identifying and communicating gaps between how AI is represented and how it is used in practice [66], but it cannot redistribute procurement authority or compel technology developers to disclose more information about how the AI in their products works. As a resource, the TARAI Index highlights relationships between product functionality and what a product claims it can do, what assumptions underpin both its functionality and claims, and how AI is represented by the product highlights transparency failures are often overlooked by existing approaches to explainability and auditing for HR tech products, yet play a central role in shaping how AI systems influence hiring decisions and integrate into the everyday practice of recruiting.

This paper contributes: (1) our conceptualization of AI transparency as a practice-based and relational phenomenon that emerges through interactions between systems, practitioners, and

¹<https://tarai.org>

organizational contexts [14, 20, 67]; (2) a database-building method for translating practitioner concerns into comparable evidence across a complex technology ecosystem; and (3) findings from analysis of 113 HR technologies showing how functionality, claims, assumptions, and AI clarity shape opacity in recruiting.

2 Related Work

2.1 From Technical to Contextual AI Transparency

AI transparency concerns whether people can understand and evaluate how an AI system is deployed and produces outcomes in context [15, 67, 69]. Generally, AI transparency is approached through interpretability and explainability: the extent to which people can understand a system's outputs or internal operation [18, 39]. These approaches often treat explainability and interpretability primarily as technical properties and assume a universal notion of transparency [28, 42, 45, 53, 59, 67]. Research on contextual [67] and social [20] transparency challenges this assumption, emphasizing that useful explanations depend on the *context where AI is deployed*, such as the domain, the end user, or the decision-making practice.

Social practice research [63, 64] directs attention beyond individual institutions to explore how practice is organized by examining the **material** elements (e.g., HR tech) through which core **competencies** of recruiting are enacted and **meaning** is created and negotiated [64]. By focusing on social practices, we can look beyond a single object, person, or system to explore the interplay of human and machine-informed decisions within a professional context over time [32, 63, 64]. In recruiting, contextual transparency depends on how practitioners interpret, adopt, and use AI.

2.2 AI in Hiring

Research on AI in hiring emphasizes bias, fairness, accountability, and potential harms. AI used in hiring can reproduce or reinforce gender and racial stereotypes [8, 12, 13, 29, 33], exacerbate structural barriers faced by people with disabilities [10, 27] and introduce evaluation practices, such as inferences from tone or biometric data that rest on problematic assumptions about job candidates [33, 44]. In response to these problems, researchers have developed methods to help audit AI used in hiring, evaluating their compliance with fairness and regulatory standards [25, 38, 66, 72]. Recent regulatory decisions and legislation at local, national, and supranational levels have classified AI used in hiring decisions as high-risk and often mandates regular auditing and oversight of such tools [1, 3, 24]. These audit-based approaches often focus on model behavior or compliance requirements, rather than how systems are integrated into everyday decision-making by recruiters at each stage of the hiring funnel.

The "hiring funnel" encompasses advertising, sourcing, screening, interviewing, background checks, and onboarding, typically involving recruiters, hiring managers, and HR personnel. Many of the HR tech products that support recruiters now feature AI integration [9, 41, 66], such as chatbots or automated skill assessments, which recruiters often feel pressured to adopt despite their limited understanding of how these systems function [57, 60, 65]. Recent research has examined how the introduction of automation into recruiting can threaten recruiters' professional discretion in candidate sourcing [65], and shows that recruiters and other HR professionals do not know how the AI works within the HR tech products they use [57, 60]. Often, the stakeholders involved in developing AI transparency disclosures for these products are not the people who commonly use such products as a part of their professional practice [62], leading to further transparency gaps. Despite these insights, there is comparatively little work examining how AI-enabled tools are embedded across the broader HR tech ecosystem, how practitioners make sense of AI across multiple systems and job processes, or how practitioners' transparency needs are shaped by their often-limited authority over HR tech product procurement.

2.3 Databases as AI Transparency and Accountability Tools

Databases have become important accountability infrastructures for documenting AI incidents, harms, and opaque automated actions [35, 43, 58]. Such databases create a collective record of AI incidents and harms to foster accountability and support ethical behavior from technology developers [43, 58]. They also support education [21], research [5], and parallel investigation in other domains, such as the use of AI-generated deepfake images for journalists [71]. Even when a “database” is not the end outcome of a research project, such as the landscape of AI accountability tools [48] or educational technology tools [31], research that classifies available tools serves a similar infrastructural purpose to database building for these outcomes.

Participation in these databases varies, ranging from submissions from the public to government-mandated reporting systems [22, 35, 43, 58]. These databases function as an infrastructural mechanism for aggregating and organizing knowledge about AI systems and focus on outcomes, such as AI failures or harms, rather than on how systems are represented, understood, and used in practice, with the closest being an examination of the potential harms of LLMs used in educational technology by teachers [31]. However, most databases are organized around harmful outcomes or regulatory compliance for researchers, regulators, or the public and offer limited support for practitioners comparing how AI is represented and used across the tools that structure their everyday work. We therefore treat database building as both an analytic method and a practice-centered intervention for contextual transparency in HR tech, particularly regarding the functionality, claims, assumptions, and clarity of AI products designed for recruiters.

2.4 Functionality, Claims, Assumptions, and Clarity in AI Systems

Embedded into all technologies are opportunities for actions that are afforded by the design of any given product [17, 19]. In recruiting, the affordances of HR tech enable the core tasks across the hiring funnel, such as candidate assessment [54], and structure data about people in similar ways [6]. Practitioners need to understand not only what these systems enable—their *functionality* [23, 65]—but also how they represent and evaluate job candidates.

To differentiate themselves to potential clients, many HR tech developers make specific claims about product features and benefits in marketing materials [66], rather than explaining how their data structures differ from competitors. Our research differentiates between actual product functionality and claims about product functionality as the because claims need not be substantiated by evidence available to practitioners. HR tech developers are not required to substantiate their **claims** about product functionality by default. Therefore, we argue that understanding a HR tech product’s use within context requires attention not only to its outcomes but also to the claims made about it and the assumptions underlying its design [66].

Claims and functionality also encode assumptions: often-unexamined beliefs about how recruiting works, what counts as candidate quality, and which attributes predict success. Examining assumptions reveals the social and technical mechanisms that make a product useful to practitioners [66]. The *normative ideas* in assumptions may clash with professional practice, making it important to identify both the assumptions and where their normative ideas come from [66]. Investigating assumptions requires understanding how technologies are used in professional practice, requiring analysis that connects product representations with practitioner accounts.

Consistency and clarity in describing AI systems and how they function are vital for fostering trust in products with embedded AI [36, 50]. End-users often seek more context around AI outcomes to improve explainability [37]. Analyzing an HR tech product’s *functionality*, *claims*, and *assumptions* can reveal the **clarity** of its AI’s description, enhancing explainability for users. Often, the details of these systems are inaccessible to users until a product license is purchased. Assessing AI clarity from a practitioner’s perspective requires understanding and distinguishing marketing claims from actual functionality and articulating the underlying assumptions that inform product design.

We analyze functionality, claims, assumptions, and AI clarity as interconnected dimensions of AI transparency. This approach allows us to identify forms of opacity extending beyond technical explainability, including inconsistencies, ambiguities, and mismatches between product and what is advertised to practitioners, and what practitioners then can do in their everyday work practice.

3 Methods: A Practice-Centered Approach to Contextual AI Transparency

We used a multi-staged, practice-centered approach combining interviews, database construction, and participatory evaluation. Database building served as both a design activity and a method for comparing how AI is represented across the HR tech ecosystem, and how transparent those representations are. Rather than treating transparency as a property of a single system, we treat it as a relational outcome that can be made legible through comparison across an ecosystem [67].

3.1 Grounding Ourselves in the Practice of Recruiting: Interview Methods

Between 2021 and 2025, we conducted over 100 semi-structured Zoom interviews with recruiters, HR managers and other hiring professions in the U.S., Canada, Australia, India, and England. These practitioner interviews captured multiple roles, industries, and work at different hiring stages. With participant consent, interviews were recorded and transcribed. In our conversations, we asked participants about recruiting workflows, technologies used, interpretations of system and AI outputs, their understandings of AI functionality, and challenges they had integrating these tools into their professional practice.

The research team independently reviewed transcripts and used inductive thematic analysis to identify recurring patterns in practitioners' experiences of AI-enabled recruiting systems. Accounts highlighted uncertainty about system behavior, distrust of outputs, gaps between vendor claims and their experiences as end-users, and threats to their professional discretion [65]. Our analysis identified practitioner-relevant dimensions such as product functionality, claims, assumptions, and clarity, that structure how AI systems are understood within the professional context of recruiting. These dimensions emerged from repeated practitioner concerns about what tools do, what vendors claim they do, what assumptions they embed, and how clearly AI use is disclosed. We center practitioners because they are responsible for interpreting and operationalizing AI-enabled hiring systems in everyday work, even when they did not select those systems or control their continued use. Their accounts therefore identify the information and interpretive burdens created by HR tech in professional practice.

3.2 Database Construction as Method

Building on interview findings, we constructed a database to examine contextual AI transparency across the ecosystem of recruiting tools. HR tech lacks centralized, standardized documentation of product functionality and AI use (see [2] for a categorization of HR tech used in various hiring stages). Further, product information is distributed across product websites, documentation, policies, audits [4], academic research [72], and practitioner accounts, making comparison near impossible. The database was designed not simply to collect that information, but to transform practitioner-defined concerns into comparable fields within the database. This comparative structure allowed us to identify recurring patterns in how AI was represented to end-users and procurement teams, how products were described and made legible across products. These patterns could not be inferred from a single product or interview alone.

3.2.1 Corpus Development. To develop a corpus of HR tech products to examine, we reviewed interview transcripts, identifying HR tech products interviewees mentioned using. We cleaned the resulting list by collapsing duplicates and removed products not specific to HR or recruiting. We also examined exhibitor lists from the annual HR Technology Conference (Las Vegas, 2021–2024), a major industry event with extensive vendor and developer representation for HR tech. Inclusion

criteria were (1) appearing on both lists, (2) multiple interview mentions, or (3) multiple conference appearances. This resulted in 113 unique HR Tech products.

3.2.2 Data Collection. We collected publicly available information from company websites, product documentation, help centers, technical/developer documentation where available, privacy and AI-policy materials, public audit reports, procurement-facing resources, and vendor materials. We supplemented organizational information with the business databases PitchBook and Crunchbase as needed. We treated “product functionality” as publicly documented functionality: the actions a product was represented as enabling through product documentation and related materials, rather than an independent verification of proprietary system behavior. Information about AI integration was often limited or inconsistent, requiring cross-referencing across sources. We cross-referenced materials using site-restricted web searches and an LLM-assisted search within specific domains, then manually verified extracted text and resolved discrepancies. Where available, we compared vendor representations with practitioner accounts, public audits, academic research, regulatory materials, and third-party documentation. These sources did not provide access to proprietary models, but they enabled us to identify inconsistencies, omissions, and divergences between advertised capability and reported use. We treated these materials, including their omissions and inconsistencies, not as neutral descriptions, but as sites where AI systems are constructed and communicated to practitioners.

3.2.3 Dimensions of the Database. The database documents HR tech products along four key dimensions: product functionality, claims, assumptions and AI clarity.

- **Product Functionality** refers to what a system does within the recruiting process, such as sourcing candidates, screening applications, or conducting technical assessments. A product’s functionality is often distributed across stages of the hiring funnel, but there are some products that support only one aspect. Functionality also categorizes AI interventions in the recruiting process. Table 1 provides examples of how AI interventions were categorized across HR tech products, including candidate elimination, assessment, and ranking functionalities.
- **Claims** refer to how systems describe their capabilities, including product descriptions, marketing language, and the values embedded into that language. These often include references to improved workflow, recruiter efficiency, automation of clerical tasks, and AI-driven decision-making and content generation.
- **Assumptions** refer to the underlying beliefs embedded in system design about how recruiters *should* do their jobs, what constitutes a qualified candidate, how candidate evaluation should occur, or which attributes of a person are predictive of success (see [33, 44, 66]). These are not always explicitly stated, but can be inferred from system outputs and features.
- **AI Clarity** refers to how clearly and consistently the use of AI within a product is communicated, including whether it is possible to determine when and how AI is involved in a product. Within the database, we operationalize this with categories such as clear, somewhat clear, unclear, or no AI.

Together, these dimensions operationalize recurring concerns raised in interviews and workshops and make those concerns comparable across products.

3.2.4 Developing the Database. The database was constructed through an iterative process of categorization and refinement of the 113 HR tech products we examined. We drew the initial categories from interview data, and they largely focused on detailing product functionality within the hiring funnel, AI use, data sourcing, and what the actual stated function of the product was. These categories were then further developed and expanded through analysis of product marketing materials and iterative design workshops with practitioners (described below).

We iteratively refined tags for hiring-stage functionality, AI interventions, vendor claims, assumptions, and AI clarity. To identify assumptions, we did not treat vendor claims as evidence of actual

AI Intervention	%(n)	Explanation	Example
Uses AI to Eliminate Ineligible Candidates	18% (21)	Included process automation and chatbots to measure candidate eligibility criteria based on job materials.	HireVue Taleo (Oracle) Jobvite
Uses AI to Assess Candidate Skill	3% (4)	Tools designed to assess candidate skill. Tools assigned candidate scores; scores were not used to advance or reject candidates.	HackerRank CodeSignal Karat
Ranks Eligible Candidates with AI	52% (58)	Products that ranked candidates in some way, sometimes after an interaction with an automated screening process. Products advertising an AI was processing information about candidates and assigning scores.	Gem iCIMS
No AI	27% (30)	Product AI did not impact candidate advancement; product had no AI.	Vetty WorkReels

Table 1. Examples of AI Intervention in HR Tech.

functionality. Instead, we compared three forms of evidence: publicly described features; vendor language about what those features were expected to accomplish; and practitioner accounts of how comparable systems were understood and used. We coded an assumption when a product feature or claim presupposed a proposition about recruiters, candidates, candidate quality, or appropriate hiring practice—for example, that automated outreach produces meaningful candidate engagement, that standardized assessments indicate job fit, or that users can appropriately interpret ranked outputs. Comparison across products then enabled us to distinguish one-off design choices from recurring assumptions embedded across the HR tech ecosystem. Table 1 summarizes the categories used to code AI interventions across products. Claims were tagged according to recurring themes such as recruiter efficiency, available automation, and claims a product was making about its AI capabilities. From these claims, the research team identified assumptions through patterns in how systems structure recruiter workflows, candidate evaluation and hiring decision-making. Finally, we documented AI clarity based on the availability, specificity, and consistency of information about AI functionality within an individual product using the following coding scheme:

- **Clear** - The product’s AI purpose and function was explicitly described. 40% (45 products)
- **Somewhat Clear** - Descriptions of AI that were contradictory or vague. 38% (43 products)
- **Unclear** - The AI’s purpose and function could not be determined. 17% (19 products)
- **No AI** - The product did not have discernible AI. 5% (6 products)

This process involved repeated comparison across HR tech products and categories by the research team, allowing us to identify patterns and inconsistencies in how HR tech products are designed and marketed to practitioners. Practitioner evaluation revealed that users could interpret AI-clarity ratings as product-quality signals rather than as narrow indicators of disclosure. We treat this as a design risk and examine it in Section 5.4.

3.3 Evaluation and Iteration on the Database

We refined the database through three rounds of iterative workshops with recruiters following established design practices [49], using the workshops to ensure the database reflected recruiting practice. Our design process began with stakeholder input, followed by prototyping and rounds of feedback and practitioner-informed refinement [16, 49]. Interview findings informed an initial prototype presented at a January 2025 public symposium at the University of Virginia. At this workshop, we asked attendees (approximately 50, primarily faculty and graduate students) to take on specific personas that use and assess HR tech products. We then conducted group and 1:1 evaluation sessions with eight HR practitioners in April–May 2025 and technology walkthroughs

with five HR executives recruited through [userinterviews.com](https://www.userinterviews.com) in July 2025. Participant job roles are reported in Appendix A, Tables 2, 3.

During each workshop session, participants engaged directly with the database to explore HR tech, compare product functionality and claims, and reflect on how AI systems align with their own decision-making processes and professional use of HR tech. For the first workshop, participants took on one of three personas and framed their analysis based around these insights. The second and third workshops were with practitioners drawing on their professional experience. At each workshop, direct interaction with the database provided insight into how transparency is interpreted, negotiated, and enacted in practice, rather than how it was presented by the researchers.

Participant feedback informed both interface design and underlying data representation, with evaluation sessions highlighting areas where additional clarity, contextual information, or improved data structure was needed within the database. These sessions also surfaced mismatches between technical descriptions of AI systems and practitioner understanding of them. This reinforced the importance of representing AI transparency in ways that align with professional practice.

3.4 Limitations

Our analysis examines the public information environment available to practitioners and procurement teams, not the proprietary models or full in-product behavior of every system. Consequently, “product functionality” in the database refers to publicly documented functionality rather than independently validated technical performance and note that there may be information that was inaccessible to the research team. We mitigated this limitation by triangulating public materials across multiple sources and comparing them with practitioner accounts where we could, but the database should not be interpreted as a definitive technical audit. Future iterations could incorporate structured contributions from practitioners with hands-on product access, including documented divergences between advertised features, licensed features, and use in organizational practice.

The team worked with English and German speakers and interviewed English-speaking practitioners, therefore, the database captures only a limited portion of the global HR tech market. Future work should examine how this analytic framework changes across non-English-language markets and with practitioners working in other linguistic, cultural, and regulatory contexts. To this end, we also did not systematically collect demographic information from all interview participants, and have taken care not to make identity-related claims because of this. Further, centering practitioners also excludes the distinct transparency needs of job candidates, and we welcome future work examining this corpus to address their transparency needs.

3.5 Methodological Contribution

Taken together, the methods outlined in this section can be reproduced as a generalizable method for studying how AI systems are embedded into the technology ecosystems of specific professional practices. Through a combination of qualitative inquiry, database design, and participatory evaluation by practitioners, this method enables the analysis of AI transparency as a relational phenomenon emerging across systems, representations, and professional practices. It can be adapted to other domains, such as educational or medical technology, where AI is embedded across multiple tools and work processes.

4 Introducing the TARAI Index

The Talent Acquisition and Recruiting AI (TARAI) Index supports contextual AI transparency in recruiting for three audiences: HR professionals who use and interpret hiring technologies; those involved in product licensing and procurement; and researchers studying AI, HR, and the future of work. For practitioners, the TARAI Index provides a way to compare product representations and identify questions to raise within their organizations. For procurement and implementation teams, it provides comparative information that may inform purchasing and renewal decisions. For

Recruiter Environment > Full Database

View All

Applicant Tracking Systems

Advertising

Sourcing

Screening

Skill Testing

Interviewing

Profiling/Background Checks

Company Name	Coun...	Ownership	Location	Company Size	Found...	Hiring...	Purpose of Tool	How the AI Works...	This Product Assumes
Workday	USA	Public	Pleasanton, CA	Enterprise	2005	ATS	Workday is a Hu...	Workday uses AI in ...	1. Assumes that HR an...
Taleo (Parent: Oracle)	USA	Public	Dublin, CA	Enterprise	1996	ATS	Taleo is an applic...	Taleo uses generativ...	1. AI algorithms can eff...

Fig. 1. The Practitioner-Oriented Environment of the Database.

researchers, it supports ecosystem-level analysis of AI interventions, vendor claims, assumptions, generative-AI features, and AI clarity across 113 HR tech products.

The TARAI Index features two interrelated views. The Recruiter Environment (see Figure 1), which allows recruiters and product procurement professions to compare products organized by hiring stage, functionality, product claims, and integrations. The Researcher Environment supports ecosystem-level analysis of AI interventions, vendor claims, assumptions, generative-AI features, and AI clarity for researchers. Together, the environments in TARAI make it possible to compare how AI is described and operationalized across the ecosystem of HR tech at each stage of the hiring funnel. Although designed around practitioner and procurement needs, TARAI is publicly accessible and has the potential to help job candidates identify technologies they encounter, although it was not designed to be a resource for job seekers.

5 Findings: AI Transparency as a Relational Outcome Across the HR Tech Ecosystem

Our method follows three linked stages. First, our practitioner interviews situate us in the professional practice of recruiting, identifying practitioners’ recurring questions, and reveal where AI-enabled systems create uncertainty and additional work. Second, the process of building the database translates those questions into comparable fields—functionality, claims, assumptions, and AI clarity—and applies them across a corpus of products. Third, practitioner evaluation tests whether those representations are intelligible, useful, or potentially misleading in practice. This three-stage sequence makes it possible to identify patterns that are difficult to observe through interviews alone, such as recurring vendor language and shared assumptions across products, while also identifying patterns that public documentation alone cannot establish, such as the interpretive labor those representations impose on recruiters.

5.1 Recruiting AI Does Not Map to Practitioners’ Information Needs

Practitioners consistently described a disconnect between AI-enabled HR tech products and their information needs in recruiting, particularly around candidate sourcing. Rather than improving their ability to identify and evaluate candidates, many recruiters felt these systems constrained access to the information they needed. P76, a technical recruiter, explained: “None of the AI tools that all these platforms have been releasing have come anywhere close to being as useful [in sourcing] as just basic Boolean operators.”² Similarly, P99, a staffing manager, described how AI-driven candidate search limited visibility of potential job candidates: ‘LinkedIn Recruiter uses a lot of AI, I can’t stand their AI because it limits the amount of information I can find. [...] I think when you just have these standard AI searches, it’s really going to limit things for recruiters.’

Database comparison showed related patterns across products. Marketing materials described similar features inconsistently and rarely specified how AI supported candidate evaluation or ranking, or how recruiters might use it. Workshop participants described inconsistent and unclear marketing language as “junk jargon” (PW1-1/P76), and observed that, as PW3-1 pointed out, some

²Boolean searching is commonly used by recruiters to narrow down candidate search results, where strings of desirable job roles, locations, professional certifications, etc. can be used with “AND” or “OR” search terms, similar to how these operators function in programming languages. 6% of the corpus used Generative AI to create Boolean search strings.

advertised features of these products “barely exist” or, as P94 explained, are fraught with technical problems. Taken together, this mismatch is not merely a usability problem for end-users, but a structural feature of how HR tech functionality is represented across the ecosystem to practitioners.

5.2 Misalignment Between Product Functionality and Claims

Participants frequently described a gap between how HR tech is marketed and products reported to function in practice. Participants described vendor claims as shaping expectations that were sometimes unmet after adoption. For example, P94, a manager at a staffing agency, explained, “*Some of those [claims made by her Applicant Tracking System] are a little misleading.*” Many HR tech products rely on broad or ambiguous descriptions of AI functionality, often collapsing distinct processes like process automation (e.g., clerical task automation, 83 distinct products, 73.5%), candidate ranking (e.g., screening for predetermined criteria 80 distinct products, 70.8%), and classification tasks (e.g., resume parsers, 32 distinct products, 28.3%) into the single category of “AI.” Across the 113 products in the database, terms such as “AI-powered” and “intelligent matching” often appeared without concrete descriptions of how that functionality worked. In one example, Textio, a product used to check for biased language in text, claimed on their website that their AI was “actually good.”

Participants described how these representations shaped their expectations and experiences using the products in question. P94 put it succinctly: “*So many people are trying to [integrate] automation[...], but there’s still so many tiny technical glitches [that] take forever to figure out. [My ATS is] not logical.*” In evaluating the database, PW1-1 noted that the plain language used to describe product functionality was useful as it cut through the “junk jargon” and empty marketing language of HR tech products. P94’s experience highlights a disconnect between product claims of improved recruiter efficiency and the realities of actually using the product, while PW1-1’s observations show the additional labor often required to interpret marketing materials in the first place. In each case, rather than reducing workload, claims experienced by participants as inaccurate or over-promising functionality often introduced more interpretive work.

Our evaluation workshops reinforced this pattern, with participants noting that it is easy to sell products that appear comprehensive, but may not actually work that well. As PW3-1 explained, products are often presented as systems that: “*‘had no bugs [and] did everything [a company] asked, and was going to make [that company] completely happy,’*” though participants questioned whether such comprehensive products exist in practice. The tension between how systems are described and how they actually function limits practitioners’ ability to meaningfully compare products and contributes to broader transparency challenges.

Further, in assessing the claims we documented in the database as a whole, several thematic clusters emerged, such as products that claimed to improve recruiter efficiency (98 products, 86.7%), automate repetitive or clerical tasks (95 products, 84.1%), and offer advanced AI capabilities (91 products, 80.5%)³, without establishing whether those capabilities were available in every product tier or performed as implied. Across the board, these recurring themes around claims create two levels of opacity: **possible overstatement**, where a product’s AI capabilities are implied but not substantiated in a meaningful way; and **under-specification**, where a product’s overall behavior is not clearly described or explained. These patterns were also reflected in practitioner observations during our evaluation sessions, where claims made by HR tech products required interpretation and validation. These findings show that AI transparency is not only a technical black-box issue, but also a phenomenon undermined by inconsistencies between how systems are described and marketed to end-users and how they operate in practice.

³Please see Appendix A for a summary table of claim clusters and categories.

5.3 Assumptions Embedded in HR Tech

Emergent from these discussions is a reflection assumptions embedded within HR tech about how recruiting should work, how candidates should be evaluated by recruiters, and how AI in HR tech should be used in practice. Assumptions operate as “taken-for-granted frameworks that guide actions without critical reflection” [32]. Rather than being explicitly stated, they are embedded in system design and surface through outputs such as rankings, classifications, and automated decisions. We identified 21 recurring assumption clusters, synthesized into four categories, that shape human–AI interaction across the hiring funnel.

5.3.1 Assumptions About How Humans Interact with HR Tech Products. The first category, 13% of observed assumptions, concerns how recruiters and candidates interact with HR tech. These include assumptions such as: “chatbots can appropriately and effectively converse with candidates” (31 products, 27.4%), and “AI or automation reduces recruiter workload or speeds up repetitive tasks” (82 products, 72.6%). These assumptions are often embedded in HR tech that automates candidate outreach across all stages of the hiring funnel. P84 valued chatbot screening it saved time: *“I can do it in 15 minutes, versus taking me 30 minutes, which gives [job candidates] more chances and [is a] better use of my time.”* Yet practitioners also described correcting automation errors and validating outputs, indicating that automation can shift rather than eliminate work (see P94 above). Rather than reducing workload or recruiter effort, HR tech products introduce additional work or add unnecessary steps to existing recruiting workflows to fix issues and determine the veracity of AI outputs. Documenting assumptions within the database allowed us to more systematically explore the gap between assumed and observed interactions with HR tech systems.

5.3.2 Assumptions About How Humans Interact with HR Tech Product Outputs. The second category, 17% of observed assumptions, concerns how people interpret, interact with, and act on HR tech outputs. These include assumptions such as: “AI can determine which candidates are the best fit for a role based on job materials” (46 products or 40.7%), and “Candidates respond positively to AI-generated messages” (44 products, 38.9%). Ranking and generative AI tools assume that recruiters can interpret and appropriately act on AI outputs, yet practitioners in our interviews reported they often could not: P29 noted that AI would rank candidates highly that did not fit the role, and *“you typically don’t have the answers”* as to why it did that, while P33 said, *“I’m just relying on the technology to do what it says it’s doing.”* These products assume a degree of interpretability and trust that participants did not consistently report. Participants also described AI-generated outreach as *“very impersonal”* (P89) and in need of tailoring (P85), particularly in relationship-driven recruiting. Analysis across the database shows these tensions reoccur, leading to distrust and additional interpretive and personalization labor for the recruiter.

5.3.3 Assumptions About How HR Tech and AI Systems Function. The third category, 35% of observed assumptions, concerns how systems are assumed to operate. These include assumptions that “AI can accurately parse resumes” (65 products, 57.5%) and “automated processes can improve recruiter effectiveness” (42 products, 37.2%). Across the database, we saw these assumptions in products used to parse candidate information for sourcing, screening, and assessment. Underpinning these assumptions was the belief that if these systems worked as described, they would improve recruiter effectiveness. For example, assessment products (e.g., CodeSignal or HackerRank) may equate standardized test performance with candidate fit, while other tools claim to infer enthusiasm or emotion from a person’s tone or word choice despite contested scientific foundations of such claims [33, 44, 66]. The recruiters we spoke to described how these assumptions shaped their evaluation processes. P80 explained: *“Our ATS [has] four options: excellent, good, bad, [or] poor answer. It would use those answers to create a score card.”* These answers, according to P80, are unclear and were not easily distinguishable. HR tech products like this reduce complex candidate profiles into standardized categories, reinforcing the assumption that candidate evaluation can

be formalized and quantified so that it is interpretable to machines. These products also assume resumes are formatted so they can be accurately parsed by an Applicant Tracking System, an assumption participants loudly contested.

5.3.4 Assumptions About How Practitioners Understand AI Outputs. The final category, also 35% of observed assumptions, concerns how practitioners are assumed to interpret AI outputs. These include assumptions such as “AI is less biased than humans and will act more fairly in hiring contexts” (50 products, 44.2%) and “humans understand AI candidate ranking and will evaluate that ranking critically” (60 products, 53.1%). Participants described a lack of transparency and calibration around AI outputs. P78 noted: *“there’s no education around it. I have no calibration of whether it’s working the way it should. There isn’t any [transparency].”* In the evaluation workshops, practitioners valued links to vendors’ AI policies and available bias or transparency audits because they clarified what was an inferred assumption in the database, rather than a vendor claim. These accounts show how assumptions about AI being “unbiased” shaped how practitioners made sense of these systems and understood their outputs without much supporting information. Understanding AI and its outputs is not just a lack of information about how AI is understood, but also a transfer of the interpretative work to the people who use these systems everyday. The database building method we took allowed us to compare technologies across the HR tech ecosystem, where that comparison showed that these assumptions were ecosystem-level patterns rather than isolated product features.

5.4 Gaps in AI Clarity and the Expanding HR Tech Market

Beyond product functionality, claims, and assumptions, participants described broader challenges in understanding when and how AI is used across systems. Our database construction showed that many HR tech products provide limited or inconsistent information about AI functionality. This lack of information makes it difficult for practitioners to determine how these products make decisions or operate generally, forcing practitioners to infer system behavior or coach their teams to disregard system features. Limited disclosure also complicates regulatory compliance, including under NYC Local Law 144 [1]. Repeatedly, practitioners described how an absence of AI clarity made it difficult for them to evaluate how a system was working, or whether the decisions it was making were biased. Although the database made located audits easier to find, those audits were not necessarily relevant to the everyday work of our participants.

These challenges are compounded by the complexity of the HR tech ecosystem and its rapid growth [55]. Practitioners described working across multiple tools and systems, each with different interfaces and assumptions, which could become overwhelming. P85, a staffing agency manager, explained: *“I’m constantly trying to follow trends and understand what tools are out there that might enhance our business.”* At the same time, organizational constraints limited practitioners’ ability to try out or adopt new HR tech. Several participants described recognizing that an alternative system might be more useful or transparent for their everyday work while acknowledging that they lacked the authority to recommend or implement such a change (e.g., P92). In contrast, those in management and decision-making positions described feeling overwhelmed by the sheer volume of HR tech products available. Within the database, being able to sort by function, hiring funnel stage, and level of AI intervention allowed practitioners, management, and procurement to identify HR tech based on business need and comparability with other HR tech products. The database provided a more comparable view of available options.

However, during the evaluation sessions, participants sometimes treated the “clear” AI clarity label as a proxy for quality rather than as an indicator of AI disclosure. W3-2 said that, if asked to identify “the top three” products for a given function, he would begin with products coded as clear; W3-4 similarly described using clear labels to narrow products for demonstration. This response is understandable in a market where information about AI functionality is often fragmented or

difficult to find, but it also creates a methodological tension: a transparency label may become an endorsement signal even when it does not measure performance, fairness, usability, or suitability for a particular organization. This finding indicates that future use of this method should distinguish AI clarity from product quality more clearly than the TARAI Index did, potentially through interface cues explicitly stating that clarity reflects the availability and consistency of public information on AI integration and transparency. User-facing guidance could also encourage comparison across functionality, claims, assumptions, and organizational fit rather than treating AI clarity as a standalone filter.

6 Discussion: Databases as Tools for Contextual AI Transparency

Our findings show that AI transparency issues in HR tech does not come solely from inaccessible AI models or from the limited knowledge of AI functionality of practitioners. Rather, opacity emerges across the relationships among product functionality in practice vs. vendor claims about functionality, how embedded assumptions shape how practitioners interact with AI and its outputs, and the professional and organizational contexts in which these systems are interpreted and used. Our database building method, informed by the transparency concerns of practitioners that we used to structure the schema of the database, allowed us to operationalize these concerns and enabled structured, cross-product comparison across 113 HR tech products. We situated the database to encourage critical reflection on the role that AI is increasingly playing in access to employment [26, 27, 57, 65, 66], where prior work has focused on transparency at the level of individual models or on particular decision points or risk areas [10, 25, 38, 72], and found that our representation of practitioner transparency concerns could be useful or misleading in practice for practitioners. Our findings outline how contextual transparency is not something that can emerge from the evaluation or auditing of a single AI model or HR tech product, but rather is a relational property of systems, how those systems represent themselves and their AI to end-users and procurement specialists, and how those tools are used and understood in practice.

6.1 Databases Offer a Fuller, but Complicated, Picture

Database comparison made ecosystem-level patterns visible that would be difficult to identify from individual products or accounts. Yet, our interviews and subsequent workshops revealed that the benefits of contextual transparency that the TARAI Index created were unevenly distributed. Many HR professionals use, interpret, and manage AI-enabled hiring systems without deciding whether or not their organizations purchase and implement them in the first place, as was often lamented by our workshop participants and interviewees. Similarly, those with procurement authority of these systems often do not work directly with the systems in question and therefore may be unaware of how transparency concerns impact those who do use these systems, as seen in our third workshop.

The TARAI Index therefore serves two interrelated purposes. It gives practitioners a structured way to communicate concerns about product functionality vs. claims, underlying assumptions, and overall clarity in how AI is described; and it gives those with procurement authority a comparative resource for acting on those concerns. While the TARAI Index was intended to be a public resource, it was designed with those doing the recruiting and hiring in mind, rather than those looking for jobs. It may help job seekers identify the HR tech they encounter while on the job market, but it was not designed around questions that may arise from transparency concerns job seekers may have. Further, the workshop findings also show that the transparency infrastructure in the TARAI Index requires safeguards against unintended product endorsement. Rating disclosure quality can be useful for narrowing a large field of products, but it should be accompanied by clear caveats that it does not assess performance, fairness, legal compliance, or organizational fit. In the version of TARAI previewed in the workshops, we did not include this as clearly as we could in the database, and we note this finding as an indication that this *should* be made explicit.

Improving transparency requires more than model explainability: it requires broad attention be paid to how systems are designed, represented to practitioners and to businesses, and how they are used. Additionally, attention needs to be paid to the assumptions that underpin the design and functionality of these systems, and how those assumptions are articulated and integrated into everyday use. This includes making assumptions more explicit, clarifying how AI is used within systems, and improving the consistency and specificity of product representations around AI, automation, and other functions of large language models, matching algorithms, and other computational tools that are subsequently collapsed into the nebulous marketing term, ‘AI’.

Still, we note that this approach is not without its limitations. The TARAI Index is time bound, captured between January and June of 2025. Even within that six month period, information changed. For example, Employ acquired Pillar, changing the information available online on both product’s websites to practitioners and subsequently shifting how those products were represented in the database. Sustaining such resources requires ongoing maintenance, but even a static database can support practice-centered transparency inquiry. We encourage further development of the method we described here, potentially adapting participatory models from the AI Incident Database and Deepfake Database [43, 71].

6.2 Marketing Language and HR Tech Opacity

Unlike incident-oriented databases [43, 71], the TARAI Index examines how vendor representations and professional practice jointly produce opacity. Our database construction method and evaluation identified recurring opacity issues across the HR tech marketplace. Marketing materials often provided incomplete or difficult-to-verify accounts of product capabilities or obscured key information such as training data and AI functionality. When that information was available, it was often buried in privacy policies or hard-to-locate documentation. These patterns warrant greater scrutiny because they limit practitioners’ access to information needed for risk-aware decisions. Contextual AI transparency in HR tech requires openly available details about product functionality and embedded AI, rather than information available only through sales or restricted channels. Future work should examine whether and how vendors solicit, incorporate, or disregard practitioner feedback after implementation. We encourage future work that explores this approach to creating contextual AI transparency within various professional settings. The method may also be adapted to other professional domains where AI is distributed across multiple tools and practices.

7 Conclusion

This paper describes how we developed the TARAI Index to study contextual AI transparency in recruiting through product functionality, vendor claims, embedded assumptions, and AI clarity. Across 113 HR technologies, opacity emerged not only from inaccessible models, but from how systems were represented, interpreted, and used within organizational practice. Database building offers a method for translating situated practitioner concerns into comparable ecosystem-level evidence, while also making visible the limits of transparency where authority, access, and information are unevenly distributed.

Acknowledgments

We’d like to thank our research participants for their insights. We’d also like to thank Michael Amadi, Sheilla Addison, and Ceilia Calhoun for their assistance at various parts of this project. Special thanks to Sarah Lebovitz and Roshni Raveendhran. This research was funded by the UVA Darden-Data Science Collaboratory Research Fellowship.

Data Availability

The current public release of the TARAI Index is available at tarai.org. The release reflects information collected between January and June 2025; vendor descriptions and product availability may have changed since that capture period.

References

- [1] 2021. New York City Local Law 144 of 2021: A Local Law to amend the administrative code of the city of New York, in relation to automated employment decision tools (AEDT). New York City Administrative Code § 20-870, enacted December 11, 2021, effective January 1, 2023. Regulates the use of automated employment decision tools in hiring and promotions; requires annual bias audits and public disclosure. <https://codelibrary.amlegal.com/codes/newyorkcity/latest/NYCAadmin/0-0-0-135598>
- [2] 2023. <https://americanstaffing.net/asa-staffing-tech-center/essential-elements-staffing-technology/>
- [3] 2024. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). Official Journal of the European Union, L, 12 July 2024, p. 1. Entered into force 1 August 2024. <https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>
- [4] ACLU. 2024. Tracking LL144 bias audits: A crowd-sourced public tracker of bias audits of Automated Employment Decision Tools (AEDTs) released by employers related to NYC's local law 144. <https://github.com/aclu-national/tracking-ll144-bias-audits?tab=readme-ov-file>
- [5] Avinash Agarwal and Manisha J Nene. 2024. Addressing AI risks in critical infrastructure: formalising the AI incident reporting process. In *2024 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECT)*. IEEE, 1–6.
- [6] Ifeoma Ajunwa and Daniel Greene. 2019. Platforms at work: Automated hiring platforms and other new intermediaries in the organization of work. In *Work and labor in the digital age*. Emerald Publishing Limited, 61–91.
- [7] Lena Armstrong, Jayne Everson, and Amy J Ko. 2023. Navigating a black box: Students' experiences and perceptions of automated hiring. In *Proceedings of the 2023 ACM Conference on International Computing Education Research-Volume 1*. 148–158.
- [8] Lena Armstrong, Abbey Liu, Stephen MacNeil, and Danaë Metaxa. 2024. The Silicon Ceiling: Auditing GPT's Race and Gender Biases in Hiring. In *Proceedings of the 4th ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization* (San Luis Potosi, Mexico) (EAAMO '24). Association for Computing Machinery, New York, NY, USA, Article 2, 18 pages. doi:10.1145/3689904.3694699
- [9] Miranda Bogen and Aaron Rieke. 2018. Help wanted: An Examination of Hiring Algorithms, Equity, and Bias. *Upturn* (2018).
- [10] Maarten Buyl, Christina Cociancig, Cristina Frattone, and Nele Roekens. 2022. Tackling Algorithmic Disability Discrimination in the Hiring Process: An Ethical, Legal and Technical Analysis. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency* (Seoul, Republic of Korea) (FAccT '22). Association for Computing Machinery, New York, NY, USA, 1071–1082. doi:10.1145/3531146.3533169
- [11] Carrie J Cai, Samantha Winter, David Steiner, Lauren Wilcox, and Michael Terry. 2019. "Hello AI": uncovering the onboarding needs of medical practitioners for human-AI collaborative decision-making. *Proceedings of the ACM on Human-computer Interaction* 3, CSCW (2019), 1–24.
- [12] Le Chen, Ruijun Ma, Anikó Hannák, and Christo Wilson. 2018. Investigating the impact of gender on rank in resume search engines. In *Proceedings of the 2018 chi conference on human factors in computing systems*. 1–14.
- [13] Phoebe K Chua, Hillary Abraham, and Melissa Mazmanian. 2021. Playing the hiring game: Class-based emotional experiences and tactics in elite hiring. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW2 (2021), 1–27.
- [14] Jennifer Cobbe, Michelle Seng Ah Lee, and Jatinder Singh. 2021. Reviewable Automated Decision-Making: A Framework for Accountable Algorithmic Systems. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (Virtual Event, Canada) (FAccT '21). Association for Computing Machinery, New York, NY, USA, 598–609. doi:10.1145/3442188.3445921
- [15] Eric Corbett and Remi Denton. 2023. Interrogating the T in FAccT. In *Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency*. 1624–1634.
- [16] Nigel Cross. 2023. *Design thinking: Understanding how designers think and work*. Bloomsbury Publishing.
- [17] Jenny L Davis. 2020. *How artifacts afford: The power and politics of everyday things*. mit Press.
- [18] Finale Doshi-Velez and Been Kim. 2017. Towards a rigorous science of interpretable machine learning. *arXiv preprint arXiv:1702.08608* (2017).
- [19] Paul Dourish. 2001. *Where the action is: the foundations of embodied interaction*. MIT press.
- [20] Upol Ehsan, Q Vera Liao, Michael Muller, Mark O Riedl, and Justin D Weisz. 2021. Expanding explainability: Towards social transparency in ai systems. In *Proceedings of the 2021 CHI conference on human factors in computing systems*. 1–19.
- [21] Michael Feffer, Nikolas Martelaro, and Hoda Heidari. 2023. The ai incident database as an educational tool to raise awareness of ai harms: A classroom exploration of efficacy, limitations, & future improvements. In *Proceedings of the 3rd ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*. 1–11.
- [22] Jana Fehr, Brian Citro, Rohit Malpani, Christoph Lippert, and Vince I Madai. 2024. A trustworthy AI reality-check: the lack of transparency of artificial intelligence products in healthcare. *Frontiers in Digital Health* 6 (2024), 1267290.

- [23] Gary W Florkowski. 2018. HR technology systems: an evidence-based approach to construct measurement. In *Research in personnel and human resources management*. Emerald Publishing Limited, 197–239.
- [24] Jon Meredith Garon. 2024. Regulation by the EEOC and the States of Algorithmic Bias in High-Risk Use Cases. *The Business Lawyer* 80, 1 (2024). https://www.americanbar.org/groups/business_law/resources/business-lawyer/2024-2025-winter/eoc-states-regulation-algorithmic-bias-high-risk/
- [25] Marissa Kumar Gerchick, Ro Encarnación, Cole Tanigawa-Lau, Lena Armstrong, Ana Gutiérrez, and Danaé Metaxa. 2025. Auditing the Audits: Lessons for Algorithmic Accountability from Local Law 144’s Bias Audits. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT ’25)*. Association for Computing Machinery, New York, NY, USA, 29–44. doi:10.1145/3715275.3732004
- [26] Nathaniel M. Glasser. 2024. *AI Hiring Tools Raise Discrimination and Privacy Concerns*. National Law Review, 2024. <https://natlawreview.com/article/what-principles-explainability-and-transparency-should-employer-consider-when-using>
- [27] Kate Glazko, Yusuf Mohammed, Ben Kosa, Venkatesh Potluri, and Jennifer Mankoff. 2024. Identifying and Improving Disability Bias in GPT-Based Resume Screening. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (Rio de Janeiro, Brazil) (FAccT ’24)*. Association for Computing Machinery, New York, NY, USA, 687–700. doi:10.1145/3630106.3658933
- [28] David Gunning, Eric Vorm, Yunyan Wang, and Matt Turek. 2021. DARPA’s explainable AI (XAI) program: A retrospective. *Authorea Preprints* (2021).
- [29] Anikó Hannák, Claudia Wagner, David Garcia, Alan Mislove, Markus Strohmaier, and Christo Wilson. 2017. Bias in online freelance marketplaces: Evidence from taskrabit and fiverr. In *Proceedings of the 2017 ACM conference on computer supported cooperative work and social computing*. 1914–1933.
- [30] Kashyap Haresamudram, Stefan Larsson, and Fredrik Heintz. 2023. Three levels of AI transparency. *Computer* 56, 2 (2023), 93–100.
- [31] Emma Harvey, Allison Koenecke, and Rene F. Kizilcec. 2025. "Don't Forget the Teachers": Towards an Educator-Centered Understanding of Harms from Large Language Models in Education. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI ’25)*. Association for Computing Machinery, New York, NY, USA, Article 1064, 19 pages. doi:10.1145/3706598.3713210
- [32] Emma Harvey, Hauke Sandhaus, Abigail Z Jacobs, Emanuel Moss, and Mona Sloane. 2024. The cadaver in the machine: The social practices of measurement and validation in motion capture technology. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–23.
- [33] Alexis Shore Ingber and Nazanin Andalibi. 2025. Emotion AI in Job Interviews: Injustice, Emotional Labor, Identity, and Privacy. In *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT ’25)*. Association for Computing Machinery, New York, NY, USA, 1–17. doi:10.1145/3715275.3732002
- [34] Alexandra Jonker, Amanda McGrath, and Alice Gomstyn. 2025. *What Is AI Transparency?* IBM. Accessed: 2025-09-09. <https://www.ibm.com/think/topics/ai-transparency>
- [35] Rishabh Kaushal, Jacob Van De Kerkhof, Catalina Goanta, Gerasimos Spanakis, and Adriana Iamnitchi. 2024. Automated Transparency: A Legal and Empirical Analysis of the Digital Services Act Transparency Database. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (Rio de Janeiro, Brazil) (FAccT ’24)*. Association for Computing Machinery, New York, NY, USA, 1121–1132. doi:10.1145/3630106.3658960
- [36] Sunnie S. Y. Kim, Jennifer Wortman Vaughan, Q. Vera Liao, Tania Lombrozo, and Olga Russakovsky. 2025. Fostering Appropriate Reliance on Large Language Models: The Role of Explanations, Sources, and Inconsistencies. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI ’25)*. Association for Computing Machinery, New York, NY, USA, Article 420, 19 pages. doi:10.1145/3706598.3714020
- [37] Sunnie S. Y. Kim, Elizabeth Anne Watkins, Olga Russakovsky, Ruth Fong, and Andrés Monroy-Hernández. 2023. "help me help the ai": Understanding how explainability can support human-ai interaction. In *proceedings of the 2023 CHI conference on human factors in computing systems*. 1–17.
- [38] Khoa Lam, Benjamin Lange, Borhane Bili-Hamelin, Jovana Davidovic, Shea Brown, and Ali Hasan. 2024. A Framework for Assurance Audits of Algorithmic Systems. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency (Rio de Janeiro, Brazil) (FAccT ’24)*. Association for Computing Machinery, New York, NY, USA, 1078–1092. doi:10.1145/3630106.3658957
- [39] Pantelis Linardatos, Vasilis Papastefanopoulos, and Sotiris Kotsiantis. 2020. Explainable ai: A review of machine learning interpretability methods. *Entropy* 23, 1 (2020), 18.
- [40] Charlotte Lytton. 2024. *AI hiring tools may be filtering out the best job applicants*. BBC Worklife, February 14, 2024. <https://www.bbc.com/worklife/article/20240214-ai-recruiting-hiring-software-bias-discrimination>
- [41] Market.us. 2024. Global AI in HR Market By Component, Deployment Mode, Technology, Application, Region and Companies - Industry Segment Outlook, Market Assessment, Competition Scenario, Trends and Forecast 2024-2033. Published April 2024; Accessed September 3, 2025. <https://market.us/report/ai-in-hr-market/>

- [42] David Martens, Galit Shmueli, Theodoros Evgeniou, Kevin Bauer, Christian Janiesch, Stefan Feuerriegel, Sebastian Gabel, Sofie Goethals, Travis Greene, Nadja Klein, et al. 2025. Beware of " explanations" of AI. *arXiv preprint arXiv:2504.06791* (2025).
- [43] Sean McGregor. 2021. Preventing repeated real world AI failures by cataloging incidents: The AI incident database. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 35. 15458–15463.
- [44] Andrew McStay. 2020. Emotional AI and EdTech: serving the public good? *Learning, Media and Technology* 45, 3 (2020), 270–283.
- [45] Tim Miller, Piers Howe, and Liz Sonenberg. 2017. Explainable AI: Beware of inmates running the asylum or: How I learnt to stop worrying and love the social and behavioural sciences. *arXiv preprint arXiv:1712.00547* (2017).
- [46] Chris Norval, Kristin Cornelius, Jennifer Cobbe, and Jatinder Singh. 2022. Disclosure by Design: Designing information disclosures to support meaningful transparency and accountability. In *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency* (Seoul, Republic of Korea) (FAccT '22). Association for Computing Machinery, New York, NY, USA, 679–690. doi:10.1145/3531146.3533133
- [47] Jessica Ochmann, Sandra Zilker, and Sven Laumer. 2021. The evaluation of the black box problem for AI-based recommendations: An interview-based study. In *International conference on wirtschafsinformatik*. Springer, 232–246.
- [48] Victor Ojewale, Ryan Steed, Briana Vecchione, Abeba Birhane, and Inioluwa Deborah Raji. 2025. Towards AI accountability infrastructure: Gaps and opportunities in AI audit tooling. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–29.
- [49] Jasper O'Leary, Holger Winnemöller, Wilmot Li, Mira Dontcheva, and Morgan Dixon. 2018. Charrette: Supporting In-Person Discussions around Iterations in User Interface Design. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–11. doi:10.1145/3173574.3174109
- [50] Hyanghee Park, Daehwan Ahn, Kartik Hosanagar, and Joonhwan Lee. 2021. Human-AI interaction in human resource management: Understanding why employees resist algorithmic evaluation at workplaces and how to mitigate burdens. In *Proceedings of the 2021 CHI conference on human factors in computing systems*. 1–15.
- [51] Hyanghee Park, Daehwan Ahn, Kartik Hosanagar, and Joonhwan Lee. 2022. Designing fair AI in human resource management: Understanding tensions surrounding algorithmic evaluation and envisioning stakeholder-centered solutions. In *Proceedings of the 2022 CHI conference on human factors in computing systems*. 1–22.
- [52] Frank Pasquale. 2015. *The black box society: The secret algorithms that control money and information*. Harvard University Press.
- [53] Forough Poursabzi-Sangdeh, Daniel G Goldstein, Jake M Hofman, Jennifer Wortman Vaughan, and Hanna Wallach. 2021. Manipulating and measuring model interpretability. In *Proceedings of the 2021 CHI conference on human factors in computing systems*. 1–52.
- [54] S Ratneshwar, Allan D Shocker, June Cotte, and Rajendra K Srivastava. 1999. Product, person, and purpose: putting the consumer back into theories of dynamic market behaviour. *Journal of Strategic Marketing* 7, 3 (1999), 191–208.
- [55] Verified Market Research. 2024. Global Talent Acquisition Software Market size By Deployment Type, By Enterprise Size, By Component By Geographic Scope And Forecast. *Verified Market Research* (2024). <https://www.verifiedmarketresearch.com/product/talent-acquisition-software-market/>
- [56] Rippling. 2024. What is HR technology? 10 types & best practices. *Rippling Blog* (2024). <https://www.rippling.com/blog/hr-technology>
- [57] Natallie Rocha. 2025. *Your Job Interviewer Is Not a Person. It's A.I.* The New York Times, July 7, 2025. <https://www.nytimes.com/2025/07/07/technology/ai-job-interviews.html>
- [58] Rowena Rodrigues, Anais Resseguier, and Nicole Santiago. 2023. When artificial intelligence fails: The emerging role of incident databases. *Pub. Governance, Admin. & Fin. L. Rev.* 8 (2023), 17.
- [59] Cynthia Rudin. 2019. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature machine intelligence* 1, 5 (2019), 206–215.
- [60] Hilke Schellmann. 2024. *The Algorithm: How AI decides who gets hired, monitored, promoted, and fired and why we need to fight back now*. Hachette UK.
- [61] Bianca Giulia Sarah Schor, Emma Kallina, Jatinder Singh, and Alan Blackwell. 2024. Meaningful Transparency for Clinicians: Operationalising HCXAI Research with Gynaecologists. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency* (Rio de Janeiro, Brazil) (FAccT '24). Association for Computing Machinery, New York, NY, USA, 1268–1281. doi:10.1145/3630106.3658971
- [62] Bianca Giulia Sarah Schor, Emma Kallina, Jatinder Singh, and Alan Blackwell. 2024. Meaningful Transparency for Clinicians: Operationalising HCXAI Research with Gynaecologists. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency* (Rio de Janeiro, Brazil) (FAccT '24). Association for Computing Machinery, New York, NY, USA, 1268–1281. doi:10.1145/3630106.3658971
- [63] Elizabeth Shove. 2007. *The design of everyday life*. Berg.

- [64] Elizabeth Shove, Matt Watson, and Mika Pantzar. 2012. *The dynamics of social practice: Everyday life and how it changes*. Sage.
- [65] Mona Sloane. 2023. Automation and recruiting: Understanding the intersection of algorithmic systems and professional discretion in the sourcing of job candidates. In *Proceedings" Automation by Design" Conference, University of Minnesota, Forthcoming*.
- [66] Mona Sloane, Emanuel Moss, and Rumman Chowdhury. 2022. A Silicon Valley love triangle: Hiring algorithms, pseudo-science, and the quest for auditability. *Patterns* 3, 2 (2022).
- [67] Mona Sloane, Ian Rene Solano-Kamaiko, Jun Yuan, Aritra Dasgupta, and Julia Stoyanovich. 2023. Introducing contextual transparency for automated decision systems. *Nature Machine Intelligence* 5, 3 (2023), 187–195.
- [68] Scott A Snell, Juani Swart, Shad Morris, and Corine Boon. 2023. The HR ecosystem: Emerging trends and a future research agenda. *Human Resource Management* 62, 1 (2023), 5–14.
- [69] Violet Turri, Katelyn Morrison, Katherine-Marie Robinson, Collin Abidi, Adam Perer, Jodi Forlizzi, and Rachel Dzombak. 2024. Transparency in the wild: Navigating transparency in a deployed ai system to broaden need-finding approaches. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*. 1494–1514.
- [70] Warren J Von Eschenbach. 2021. Transparency and the black box problem: Why we do not trust AI. *Philosophy & technology* 34, 4 (2021), 1607–1622.
- [71] Christina P Walker, Daniel S Schiff, and Kaylyn Jackson Schiff. 2024. Merging AI incidents research with political misinformation research: introducing the political Deepfakes incidents database. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 23053–23058.
- [72] Christo Wilson, Avijit Ghosh, Shan Jiang, Alan Mislove, Lewis Baker, Janelle Szary, Kelly Trindel, and Frida Polli. 2021. Building and Auditing Fair Algorithms: A Case Study in Candidate Screening. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (Virtual Event, Canada) (FAccT '21). Association for Computing Machinery, New York, NY, USA, 666–677. doi:10.1145/3442188.3445928
- [73] Lucas Wright, Roxana Mika Muenster, Briana Vecchione, Tianyao Qu, Pika Cai, Alan Smith, Comm 2450 Student Investigators, Jacob Metcalf, J Nathan Matias, et al. 2024. Null Compliance: NYC Local Law 144 and the challenges of algorithm accountability. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency*. 1701–1713.

A **Appendix A: Tables & Figures**

P#	Workshop Session	Participant & Job Role
PW1-1 (P76)	1	PW1-1. Senior Technical Recruiter
PW2-1 PW2-2 PW2-3 PW2-4	2	PW2-1. Managing Director PW2-2. Senior Campus Recruiter PW2-3. Recruitment Manager PW2-4. Talent Acquisition Recruiter
PW3-1	3	PW3-1. Technology Developer
PW4-1	4	PW4-1. HR Influencer
PW5-1	5	PW5-1. Chief Risk & Compliance Officer

Table 2. Participant job roles for the practitioner workshops.

P#	Job Role	Age	Reported Gender
W3-1	[Anonymized HRIS] Operations Manager	62	Woman
W3-2	HR Director	54	Man
W3-3	Early Careers Leader	53	Woman
W3-4	Human Resources Director	55	Woman
W3-5	VP Human Resources	55	Woman

Table 3. Participant job roles for the interviews with Senior HR Professionals in Workshop 3.

Claim cluster	Claim Categories	# Distinct products (%)
Improved recruiter efficiency	Process Automation for Clerical Tasks; Generative AI for Clerical Tasks; Process Automation to Find Candidates; Generative AI to Create Boolean Search Strings; Chatbots to Interview Candidates; Process Automation to Interview Candidates	98 (86.7%)
Automation of repetitive or clerical tasks	Process Automation for Clerical Tasks; Generative AI for Clerical Tasks; Process Automation to Find Candidates; Process Automation to Interview Candidates; Chatbots to Interview Candidates	95 (84.1%)
Advanced AI capabilities	AI to Evaluate and Rank Candidates; AI to Screen Candidates for Pre-Determined Criteria; AI to Screen Candidates for Diversity Indicators; AI to Conduct Sentiment Analysis; AI to Evaluate Candidate Behavior and Tailor Questions; Search Algorithms to Parse Resumes; Generative AI to Reduce Bias in Hiring; Generative AI to Create Boolean Search Strings	91 (80.5%)

Table 4. Claim clusters and foundational claims.